

Prarthana Krishnamurthy

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Summary

Machine Learning & Data Engineer with 4+ years of experience bridging advanced AI/generative models with molecular modeling (Chemprop, RDKit) and production data pipelines. Proven expertise in deploying agentic tools (MCP, FastAPI), PySpark ETL, and full-stack solutions (React, Docker/CI/CD) to solve complex scientific workflows.

Experience

Machine Learning Intern | *Molecular Modeling*

May 2026 – Jul 2026

Alkermes Inc.

Waltham, MA

- Built supervised ML models (PyTorch, scikit-learn) predicting compound target properties from tabular assay data, achieving an average R^2 of 0.7 and Spearman correlation of 0.73 for lead candidate prioritization
- Deployed a FastAPI serving layer for the molecular property models, cutting inference latency 2.5x, and exposed it as both a REST API and an MCP layer for agent-callable scientific workflows
- Established the team's ML deployment infrastructure (Docker, GitLab CI/CD), making it a production-ready pipeline
- Applied SHAP to translate black-box model outputs into explanations for scientists and non-technical stakeholders
- Developed the data processing and feature-engineering pipeline (RDKit) for the compound property prediction models, computing molecular descriptors and fingerprints and cleaning the underlying assay dataset

Machine Learning Research Assistant

Jan 2025 – Apr 2026

Northeastern University (Advisor: Prof. Dino Konstantopoulos)

Boston, MA

- Architected transformer and diffusion-based generative models, leveraging flow matching to address complex non-autoregressive text generation constraints
- Scaled training across 8x NVIDIA H100 GPUs via PyTorch FSDP and NCCL optimization, sustaining 95%+ GPU utilization
- Reduced diffusion validation loss by 92.3%, achieving a 0.93 BERT semantic similarity score
- Resolved sampling error accumulation by implementing DPM-Solver++, reducing discretization error to 0.65
- Delivered SOTA performance metrics including BLEU-4: 0.267 and ROUGE-L: 0.742, demonstrating end-to-end ownership

Software Engineer, Data

Sep 2021 – Dec 2023

Societe Generale Global Solution Centre

Bangalore, KA

- Optimized ETL pipelines to automate raw data ingestion, delivering high-quality, structured datasets
- Partnered with data scientists to design Parquet schemas, unifying multi-format data within a central data lake
- Engineered PySpark transformation pipelines for raw-to-silver layer processing, performing large-scale data wrangling, deduplication, schema enforcement, and standardization across multiple data sources
- Refactored 50+ legacy scripts into Airflow DAGs, enabling scalable workflow orchestration and centralized monitoring
- Productionized 3+ Dockerized enterprise applications on Azure Kubernetes Service, processing 1M+ daily transactions

Software Engineering Intern

Mar 2021 – May 2021

Capgemini

Bangalore, KA

- Designed and built 15+ RESTful microservices using Java 17 and Spring Boot, reducing API latency by 35% through optimized Hibernate JPA caching
- Implemented scalable multi-tenant microservices, ensuring high availability and data consistency for enterprise clients
- Built RESTful APIs, optimizing response times through efficient data handling and database query optimization
- Engineered unit and integration test suites (JUnit, Mockito) to catch regressions and ensure API reliability
- Developed a client-facing UI in React on top of the microservices, exposing backend services and data directly to enterprise clients

Publications

- [Non-Autoregressive Text Generation Using Diffusion Models](#), Master's Thesis, Northeastern University

Technical Skills

Programming Languages: Python (NumPy, SciPy, Pandas, Matplotlib), SQL, Java, React, Bash, Linux

Cheminformatics: RDKit, Chemprop, PyMol, DeepChem

ML/DL Frameworks: PyTorch (FSDP, DDP), TensorFlow, Keras, scikit-learn, XGBoost, LightGBM

Deep Learning & Generative AI: Transformers, Diffusion Models, Flow Matching, Optuna, LLMs, RAG

Cloud, MLOps & Deployment: AWS (S3, ECS, Lambda, Kinesis, DynamoDB, CloudWatch, CloudFormation), Azure (Data Lake, Kubernetes Service), Docker, Kubernetes, Terraform, FastAPI, MCP, REST APIs, Microservices, Git, CI/CD

Data Engineering: Airflow, PySpark, Elasticsearch, ETL/ELT Pipelines, Parquet, Kafka, Apache Flink, Hadoop

Education

Northeastern University, MS in Computer Software Engineering (GPA: 3.84/4.0)

Jan 2024 – Dec 2025

Vishvesvaraya Technological University, BE in Electronics and Communication

Aug 2017 – Aug 2021